

Trying a cumulative-potential readout on matched SHD networks

exp071 · 19 July 2026 · Draft

Abstract

This one-seed exploratory experiment tests whether the newly merged `tools/snn` cumulative-potential decoder can move matched Dale-constrained COBA and PING networks above the exp068–exp070 validation plateau on SHD. The screening ladder uses the same deterministic development-training / held-out-validation split as exp069 and exp070; the official SHD test remains sealed. Candidate one keeps the baseline 256-cell architecture and changes the classifier to `--readout cumulative-potential --signed-readout`

`--readout-bias`. If that is finite, active, and promising, candidate two keeps the same decoder and tests two 256-cell hidden layers. Only the best short candidate may be promoted to forty epochs.

Goal prompt

```
/goal Design and execute the next exploratory SHD experiment on branch night/spiking-heidelberg-digits/ar071, using the newly merged tools/snn cumulative-potential readout additions to try to raise validation accuracy while keeping the comparison between matched Dale-constrained COBA and PING networks scientifically clean.
```

```
Start from updated main at merge commit
9527162d7a0ef80deac0e3f606ce513b9280ba6c. Do not modify main. Open a new
draft PR for this experiment when the first meaningful commit is ready.
```

Scientific objective:

Run a one-seed exploratory validation-only ladder on SHD to identify whether stronger temporal readout and modest capacity changes materially improve matched COBA/PING learning beyond exp068-exp070.

Constraints:

- Use one seed only.
- Do not use the official SHD test set during exploratory screening.
- Use a held-out validation split from the training set for all selection decisions.
- Keep COBA and PING matched except for their registered cell-specific voltage-gradient dampening.
- Preserve matched input, readout, training split, optimizer, batch, and preprocessing settings across COBA and PING within each candidate.
- Use short runs first for iteration speed, targeting about 8-10 epochs per

candidate.

- Promote only the most promising candidate to a 40-epoch validation run.
- Do not run multiple seeds unless a later claim is worth defending and I explicitly authorize it.
- RunPod spending requires explicit approval before pod creation; use at most one pod at a time unless separately authorized.
- Reap any pod when finished.
- Do not merge the PR.
- Avoid the full local test suite on the 4 GB Hetzner host; use focused checks and demolab build.

Candidate ladder:

1. Baseline architecture with the new signed cumulative-potential readout:
--readout cumulative-potential --signed-readout --readout-bias
2. If candidate 1 is finite, active, and improves validation learning, try modest increased capacity with two hidden layers, e.g. --n-hidden 256 256, keeping the same readout.
3. Optionally test one conservative recurrence/readout-adjacent setting if supported by existing tools/snn CLI and scientifically justified before seeing final results.
4. Promote the best candidate, if any, to a matched 40-epoch COBA/PING validation run.

Deliverables:

- Create the next numbered experiment, likely exp071, with artifacts under artifacts/data/exp071.
- Produce numbers.json, provenance, reproducer, training curves, firing-rate diagnostics, and matched input/E/I rasters where relevant.
- Write writings/exp071.typ as the canonical cold-readable experiment report.
- Include the goal prompt below the abstract and append timestamped activity-log/thread checkpoints in the experiment appendix, following the simplified SHD organization.
- Record scientific decisions, failures, anomalies, costs, commits, results, and pending work in the experiment article.
- Sanitize any publishable transcript/log content before committing.
- Build and validate with focused checks plus demolab build.
- Commit meaningful attempts, including killed attempts, with clear messages.
- Push branch night/spiking-heidelberg-digits/ar071 and update the new PR with evidence.
- Finish at the human review gate with a concise evidence summary, exact compute spend, validation performed, and links to the rendered exp071 file and PR.

Methods

Locked comparison

Both cells use seed 42, the exp069/exp070 development split (7340 training utterances and 816 validation utterances), batch size 32, Adam learning rate 0.0004, 1 ms simulation steps, 1000 ms utterance windows, fixed Dale-constrained feed-forward/recurrent conductances, and

identical input preprocessing. COBA disables the inhibitory loop and uses the engine’s no-dampening value; PING enables the inhibitory loop and uses voltage-gradient dampening 1000. These cell-specific settings are the only intended COBA/PING difference inside each candidate.

The readout is the new cumulative-potential decoder with signed abstract classifier weights and a trainable readout bias. The readout initialisation scale is one, not the old membrane-readout scale of 225, because that scale belonged to the prior output-LIF membrane recipe. COBA and PING are still matched exactly within each candidate.

Ordered exploratory ladder

Order	Candidate	Short epochs	Status
1	Baseline 256-cell architecture plus cumulative-potential readout	8	registered
2	Two hidden layers, 256 cells each, same readout	8	conditional

Candidate selection uses held-out validation accuracy and cross-entropy only. Training must remain finite, active, non-saturated, and free of skipped or non-finite updates. The official test file is not staged or loaded by the runner.

Current evidence

No paid compute or full screening run has started in this article state. The artifact records spend as 0 USD and active pods as 0. The local 128/128 two-epoch smoke passed: COBA selected 5.47% with final E rate 24.25 Hz, and PING selected 10.16% with final E/I rates 2.98 / 15.05 Hz. These smoke accuracies are plumbing diagnostics, not screening evidence. Pending work is the first commit/PR checkpoint, explicit RunPod spending gate, short candidate pair, and promotion decision.

Activity appendix

Checkpoint cp001

Timestamp: `#trace.checkpoint_time_utc`. Sanitized source hash prefix: `#trace.sanitized_sha256_prefix`. The log excludes hidden reasoning, tool payloads, credentials, private paths, environment values, addresses, and sensitive infrastructure details.

Decisions, actions, and pending work

1. The experiment is registered as exp071 on branch night/spiking-heidelberg-digits/ar071 from main merge commit 9527162d7a0ef80deac0e3f606ce513b9280ba6c.
2. Candidate one is fixed before results as the baseline 256-cell architecture with cumulative-potential readout, signed abstract classifier weights, and readout bias.
3. The old membrane-readout initialisation scale of 225 is not carried into the cumulative decoder; readout scale is one for the new decoder, matched across COBA and PING.
4. Candidate two is conditional on candidate-one viability and keeps the same readout while using two 256-cell hidden layers.
5. Selection remains one-seed and validation-only on the exp069/exp070 split. The official SHD test remains unavailable to the runner.
6. No RunPod pod has been created in this checkpoint and spend remains 0 USD.

Visible messages

User

2026-07-19T11:18:05.855Z

Role: user · Checkpoint: cp001 · Session: exp071-continuation-redacted

<codex_internal_context source="goal"> Continue working toward the active thread goal: design and execute the next exploratory SHD experiment on branch night/spiking-heidelberg-digits/ar071 using the newly merged cumulative-potential readout additions, with one seed, validation-only selection, local short runs before any promoted forty-epoch run, no official test use, explicit RunPod approval before spending, focused checks, artifacts under artifacts/data/exp071, writings/exp071.typ, timestamped activity checkpoints, commits, push, PR update, and no merge. </codex_internal_context>

Assistant

2026-07-19T11:18:05.855Z

Role: assistant · Checkpoint: cp001 · Session: exp071-continuation-redacted

I'll pick this up from the actual repo state, keep the dirty article-consolidation work isolated, and start by setting up the exp071 branch/work area cleanly. First I'm running the lab manual hook, then I'll inspect what's already in place.

Assistant

2026-07-19T11:18:05.855Z

Role: assistant · Checkpoint: cp001 · Session: exp071-continuation-redacted

Because this goal includes opening/updating a PR, I'm using the GitHub workflow skill now; it'll keep the GitHub side tidy and bounded.

Assistant

2026-07-19T11:18:05.855Z

Role: assistant · Checkpoint: cp001 · Session: exp071-continuation-redacted

The main checkout has the older article-consolidation changes restored and dirty, so I'm not touching it. I'm making a separate Git worktree for night/spiking-heidelberg-digits/ar071; that gives exp071 a clean branch without risking those staged/unstaged edits.

Assistant

2026-07-19T11:18:05.855Z

Role: assistant · Checkpoint: cp001 · Session: exp071-continuation-redacted

I've got the design shape now: exp071 can inherit exp069's split, selection, validation-only test seal, rasters, and RunPod plumbing, while overriding only the candidate-specific readout/hidden sizes/epochs. I'm going to add the initial scaffold plus a pre-result artifact so the report builds before paid compute.

Assistant

2026-07-19T11:18:05.855Z

Role: assistant · Checkpoint: cp001 · Session: exp071-continuation-redacted

One design decision I'm making explicit before results: candidate 1 uses the cumulative decoder with `--signed-readout` `--readout-bias` and a readout init scale of 1.0, not the old membrane-readout scale of 225. That scale was

specific to the previous output-LIF membrane recipe; keeping it would be an unregistered hybrid rather than the clean new readout test. COBA/PING remain matched within each candidate.

Checkpoint cp002

Timestamp: #trace-two.checkpoint_time_utc. Sanitized source hash prefix: #trace-two.sanitized_sha256_prefix.

Decisions, actions, and pending work

1. Focused Python compilation passed for experiments/exp071.py.
2. Focused Ruff initially caught an unused import in experiments/exp071.py; the import was removed and Ruff then passed.
3. RunPod dispatch was dry-run only. It planned two 5090 pods, one for COBA and one for PING, and explicitly created nothing.
4. The first local smoke trained both cells finite and active, but the runner failed its own added hidden-size validation because hidden_sizes is stored in config.json while metrics.config keeps only legacy n_hidden.
5. The validation check was fixed to read config.json for hidden_sizes, falling back to n_hidden only when needed.
6. The second local 128/128 two-epoch smoke passed for both cells with the cumulative-potential readout. COBA selected 5.46875% validation accuracy and ended at 24.25390625 Hz E; PING selected 10.15625% and ended at 2.9765625 Hz E / 15.046875 Hz I.
7. The smoke accuracies are plumbing diagnostics only, not registered screening evidence.
8. No paid compute has started and spend remains 0 USD.

Visible messages added

Assistant

2026-07-19T11:28:51.466Z

Role: assistant · Checkpoint: cp002 · Session: exp071-continuation-redacted

Pre-result artifacts published locally. I'm doing the light checks in increasing cost: Python compile, dry-run RunPod dispatch (no -live), then the local 128/128 two-epoch smoke if the cheap checks pass.

Assistant

2026-07-19T11:28:51.466Z

Role: assistant · Checkpoint: cp002 · Session: exp071-continuation-redacted

Cheap checks found one real scaffold nit—an unused import—while the RunPod path correctly dry-ran two pods and created nothing. I'm cleaning that import and then running the local smoke.

Assistant

2026-07-19T11:28:51.466Z

Role: assistant · Checkpoint: cp002 · Session: exp071-continuation-redacted

The smoke exposed a useful plumbing bug, not a science failure: both cells trained finite/active, but my extra validation looked for hidden_sizes inside metrics.config, where this engine build stores only the legacy n_hidden.

I'm tightening that check against config.json instead and recording this as a killed/debugging attempt before moving on.

Assistant

2026-07-19T11:28:51.466Z

Role: assistant · Checkpoint: cp002 · Session: exp071-continuation-redacted

Ruff and compile are clean now. Starting the second local smoke; still no cloud spend.

Assistant

2026-07-19T11:28:51.466Z

Role: assistant · Checkpoint: cp002 · Session: exp071-continuation-redacted

Local smoke passed: COBA and PING were both finite and active on the 128/128, 2-epoch gate. I'm promoting that from temp into the publishable pre-result artifact and adding a second timestamped checkpoint before build/commit.

Assistant

2026-07-19T11:28:51.466Z

Role: assistant · Checkpoint: cp002 · Session: exp071-continuation-redacted

Build caught a subtle artifact-publishing issue: my pre-result publish swapped the exp071 artifact directory and dropped the activity JSON files I'd just written. The site build protected itself by stubbing exp071. I'm fixing the artifact layout by restoring those JSON logs after publish and then rebuilding.